SHANA IBATUAN

atejanelle18@gmail.com | 310-400-9118 | cherryonntopp.github.io/Portfolio/ | linkedin.com/in/shana-ibatuan

EDUCATION

University of California, San Diego – Jacobs School of Engineering La Jolla, CA
Bachelor of Science in Computer Science Expected June 2029

- **Coursework:** Introduction to Programming and Computational Problem Solving, Calculus for Science and Engineering (Integral calculus, series, applied problem-solving), Discrete Mathematics, Data Structures and Object-Oriented Design, Systems Programming and Software Tools
- **Involvement:** Association for Computing Machinery (ACM), Women in Computing (WiC)

California Academy of Mathematics and Science *High School Diploma* Carson, CA

SKILLS

- **Languages:** Python, Java (OOP), JavaScript, HTML/CSS, Tailwind CSS, MIT App Inventor
- **Frameworks/Tools:** Colab, TensorFlow, Flask, Git, GitHub, VS Code,

EXPERIENCE

Summer Program for Incoming Students @ UCSD | *Participant* August 2025 - September 2025

- Completed an intensive 5-week residential program focused on the foundations of computer science, algorithmic problem-solving, and mathematical modeling.
- Brain Tumor MRI Classifier: Developed a Convolutional Neural Network (CNN) using TensorFlow and Keras to classify MRI scans, optimizing model architecture to increase diagnostic accuracy from 74% to 93%.
- Collaborated with peers to implement and debug complex algorithms on Raspberry Pi hardware.

NASA STEM Enhancement in Earth Science | *Research Intern* July 2024 - August 2024

- Designed and optimized a 3D-printable lunar rover wheel using FEA in Fusion 360, achieving high compliance while maintaining a safety factor above 3 under realistic load constraints
- Automated analysis of simulation results by aggregating FEA outputs and processing data in MATLAB

STEM Active Learning Program @ CSU Dominguez Hills | *AI Intern* June 2023 - August 2023

- Built and trained a convolutional neural network in TensorFlow/Keras for image classification
- Improved model accuracy from ~74% to ~93% by tuning architectures and hyperparameters, using preprocessing and feature extraction layers
- Standardized training pipelines through data augmentation and normalization, comparing performance across three training approaches

PROJECTS

BrainScan AI - Medical Imaging/Brain Tumor Classification | *Developer* Summer 2025

- Created a convolutional neural network (CNN) in Python using TensorFlow to classify brain tumors (meningioma, pituitary, glioma, and no tumor) from MRI images, achieving a 94% testing accuracy
- Strengthened Python skills by organizing code into clear modules, working with arrays and image data, and applying functions and conditions
- Developed a web interface using Flask, HTML, Tailwind CSS, and JavaScript to interact with the CNN model for MRI uploads and model predictions. integrating backend with a clean frontend
- Received NCWIT Collegiate Finalist Award for this project

HONORS AND AWARDS

NCWIT Aspirations in Computing Collegiate Award Finalist Issued by NCWIT 2025
Regents' Scholarship Issued by UC San Diego 2025